

Comparison of Ultrasound-Hot Plate and High-Shear Centrifugal Mixing Techniques for Fabricating CNT-Reinforced Nanocomposite Panels

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1. Introduction

- Composite materials are often used to fabricate high-performance structures, such as aircraft fuselage skin, frames, and wings.
- Laminated composites have weaker interlaminar properties than in-plane properties because they are held together only by the epoxy matrix in the thickness direction. Carbon Nanotubes (CNTs) are used as reinforcements within the resin to improve overall composite strength.
- A traditional technique to integrate CNTs within an epoxy matrix is to utilize ultrasonication and a hot plate to evaporate a solvent, shown in Figure 1. This process is time-consuming and labor-intensive.

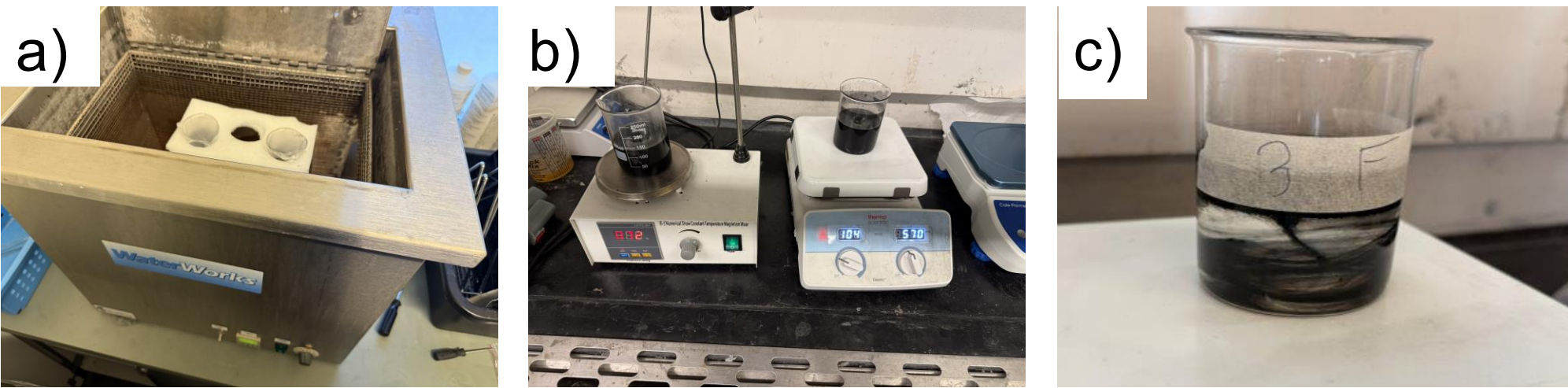


Figure 1. Nanocomposite manufacturing equipment for ultrasound-hot plate method: (a) sonicating CNTs in ethanol, (b) mixing CNTs with epoxy resin on hot plate, and (c) epoxy-CNT mixture close-up.

2. Objective

- The main goal of this research was to study the influence of high-shear centrifugal mixing techniques on the mechanical properties of CNT-reinforced nanocomposite panels using open-hole tensile testing, as per ASTM D5766/D5766M – 23. This research aims to compare these results to those obtained for similar panels manufactured with the traditional ultrasound-hot plate method.

3. Fabrication

- The research used EPON 815C epoxy resin and EPIKURE 3282 hardener. Multi-walled straight, helical, and functionalized straight CNTs were studied (MWSCNTs, MWHCNTs, and OH-FMWSCNTs, respectively). For each CNT type, two CNT-epoxy panels were fabricated with (1) 0.025 wt% and (2) 0.050 wt% CNT reinforcement.
- A schematic of the test coupon per the ASTM standard is shown in Figure 2. The fabrication steps are shown in Figure 3.

3. Fabrication (cont'd)

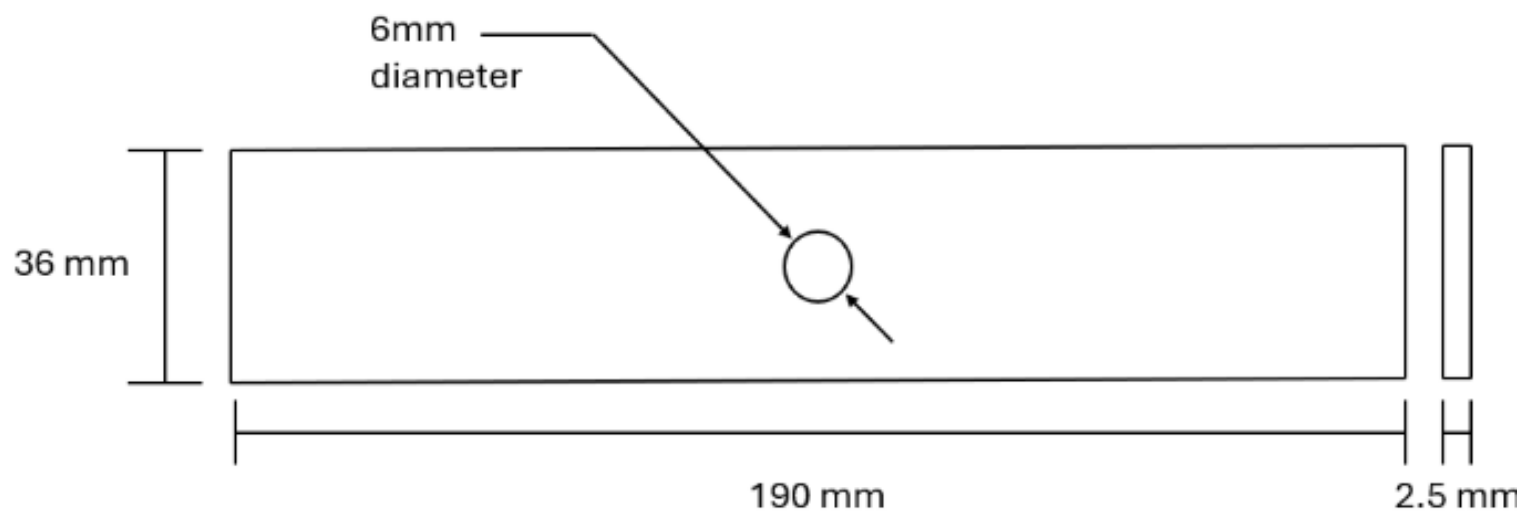


Figure 2. Test specimen dimensions per ASTM D5766/D5766M – 23 requirements.

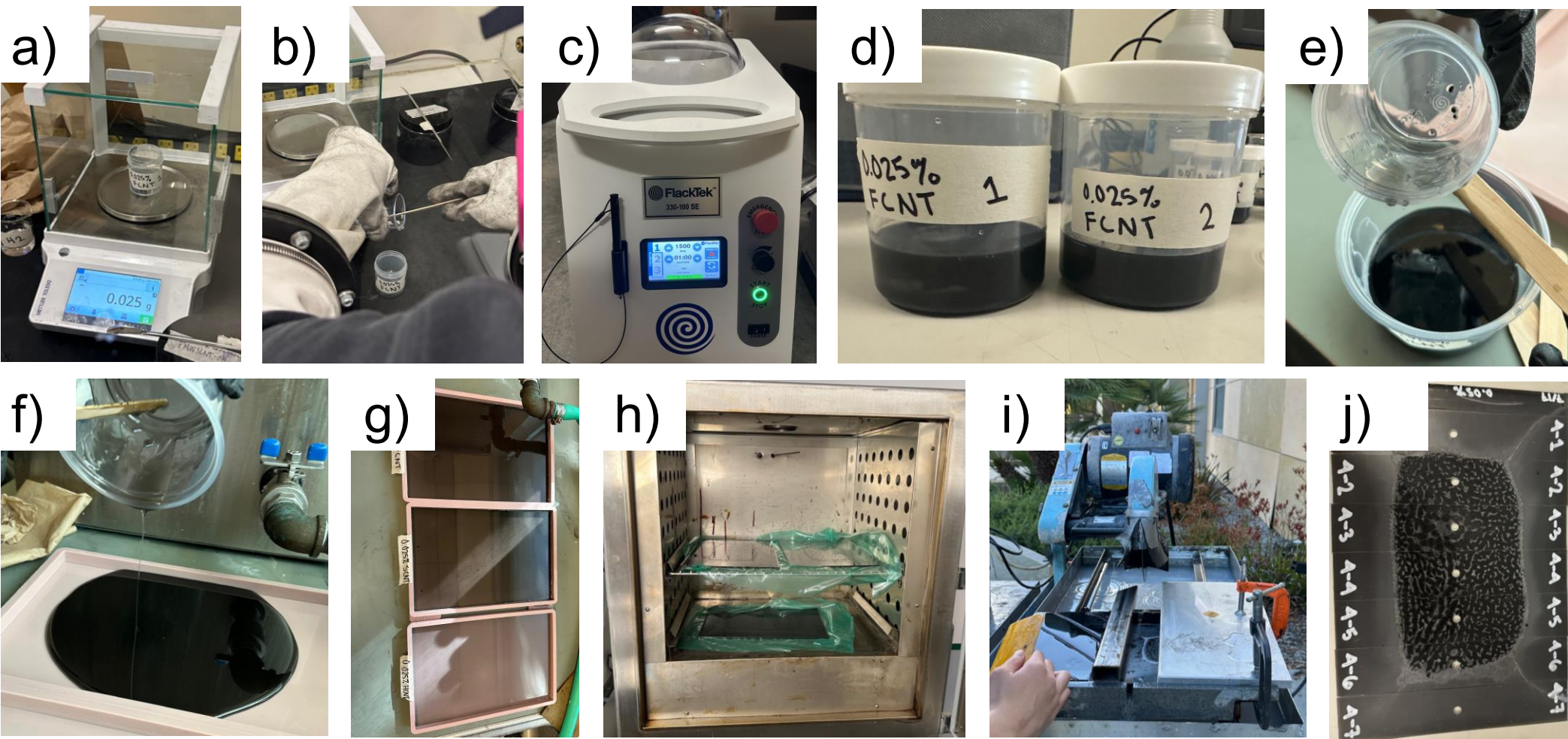


Figure 3. Nanocomposite specimen fabrication process: (a) weighing CNTs on precision scale, (b) combining epoxy with CNTs, (c) performing high-shear mixing, (d) resulting CNT-epoxy mixtures in separate containers (e) combining mixture with hardener, (f) pouring panel into rubber mold, (g) leaving poured panels to cure overnight, (h) post-curing panels in industrial oven, (i) cutting coupons to size on wet tile saw, and (j) drilling and reaming centered holes in each specimen.

4. Testing

- Figure 4 indicates the testing setup utilized for the open-hole tensile tests and shows testing specimens post-failure.

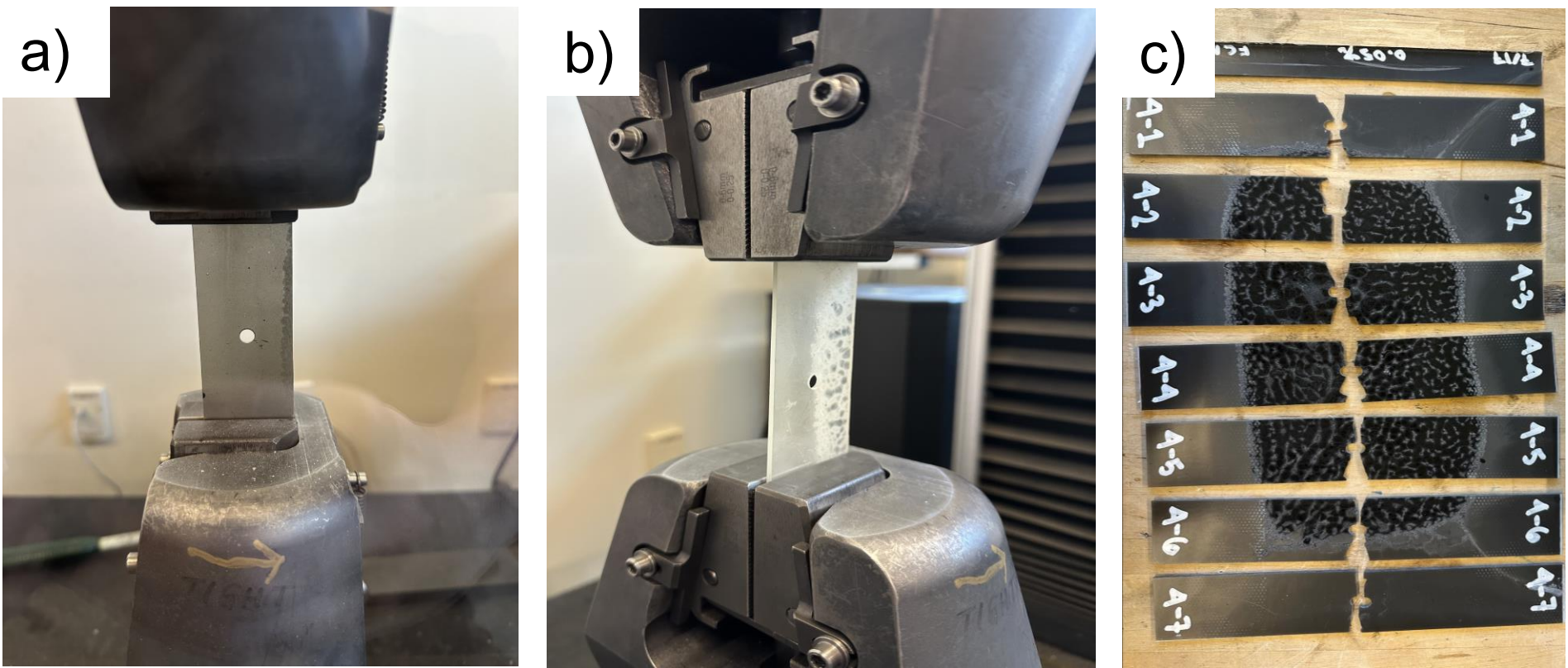


Figure 4. Open-hole tensile test setup: (a) tensile fixture orientation front view, (b) tensile fixture orientation angled view, and (c) tensile specimens' post-failure.

5. Results and Discussion

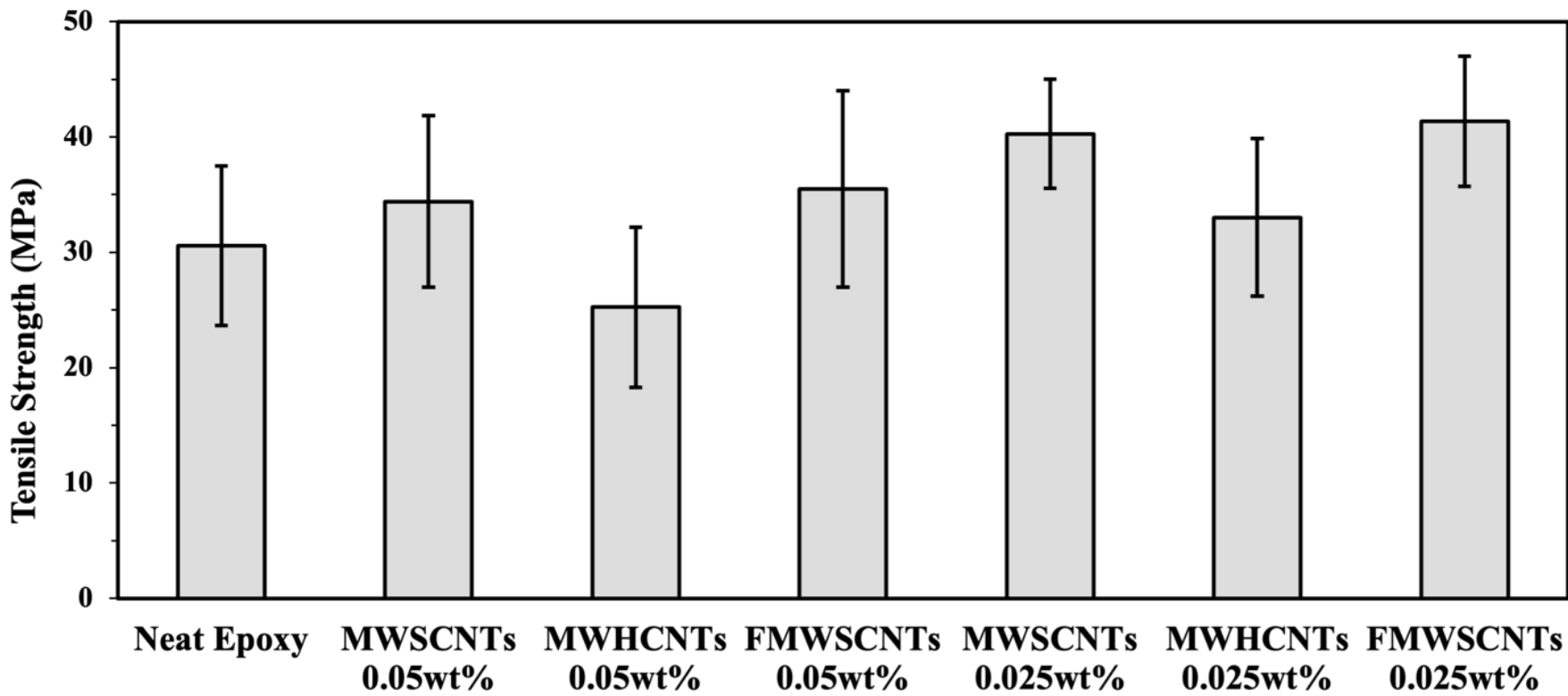


Figure 5. Tensile strength comparison between shear-mixed samples and neat epoxy.

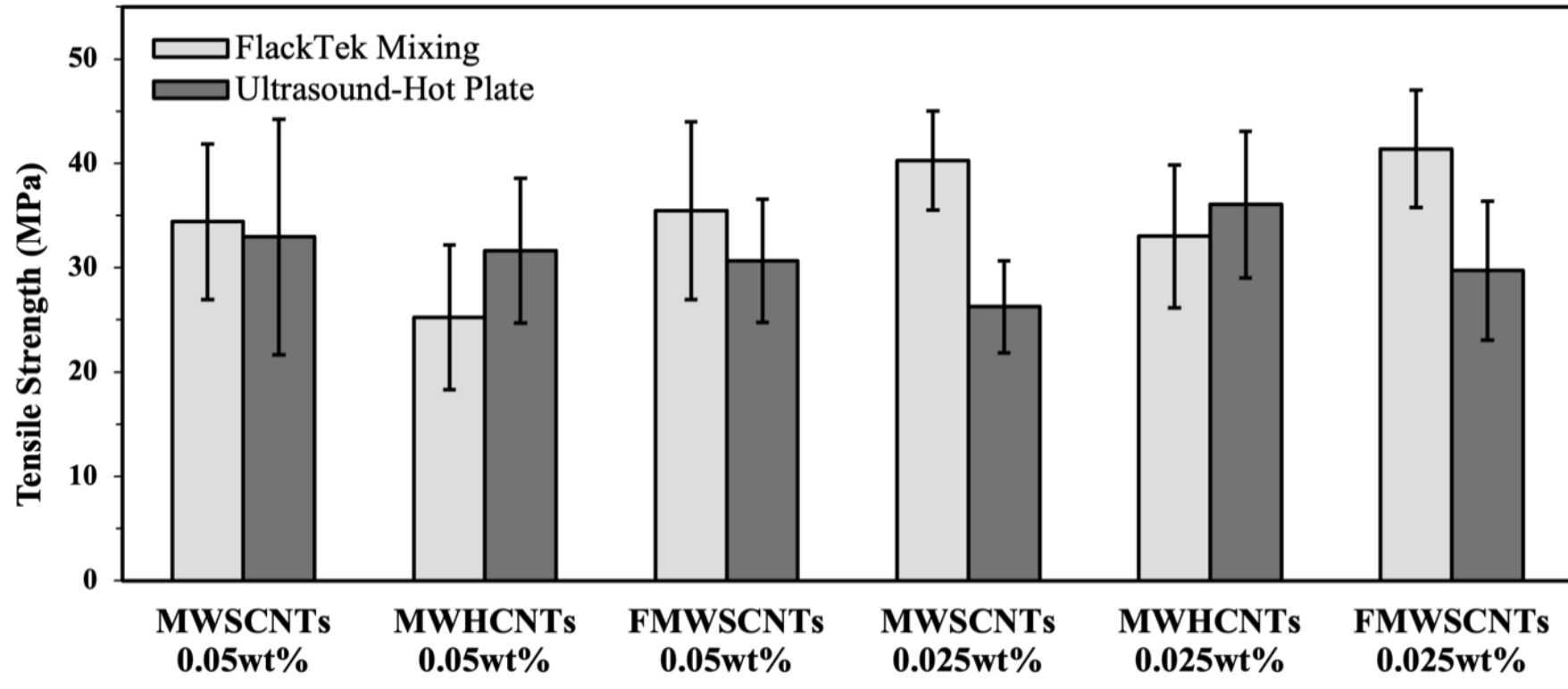


Figure 6. Tensile strength comparison between high-shear samples and ultrasound-hot plate samples.

6. Conclusion

Table 1. Differences in mechanical properties between the two fabrication methods for nanocomposite test samples (high-shear mixing results minus ultrasound-hot plate results).

Sample	Tensile Strength Difference [%]	Tensile Modulus Difference [%]	Strain Difference [%]
MWSCNT 0.025 wt%	53.4	-16.6	50.9
MWHCNT 0.025 wt%	-8.4	-7.5	2.3
FMWSCNT 0.025 wt%	39.2	-5.0	7.1
MWSCNT 0.05 wt%	4.4	0.3	-21.3
MWHCNT 0.05 wt%	-20.2	-7.7	-16.2
FMWSCNT 0.05 wt%	15.7	-5.7	13.4

7. References

- R. Sritharan, C. Bentley, and L. Granillo, "Enhancement of the Open Hole Tension Properties of the Nanocomposites using Carbon Nano-Coils," presented at the CAMX 2025.
- R. Sritharan and A. Davood, "Enhancement of the Mechanical Properties And Bonding of Composite Laminates and Assemblies Using Carbon Nano-Coils," presented at the CAMX 2022.